BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLY PROGRESS REPORT (22/06/2026 – 28/06/2026)

WEEK 6

Project Name	OSINT Collection from Social Media Network
PROJECT DESCRIPTION :	Continuation of the linkmap OSINT pipeline. Week 5 focused on finalising and stabilising the Nexus graph explorer with two major additions: a date-range handle filter with multi-select handle picker, and an automated PDF OSINT report generator that compiles per-handle intelligence profiles, cross-handle relationship analysis, and narrative inferences into a downloadable report. The week also fixed a critical SPA routing bug that was intercepting file downloads.
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WEEK 6 — DAILY WORK LOG

Day 1–2 | 22–23 June 2026 | Autocomplete Handle Search — Design & Implementation

The first two days of the week were dedicated to designing and building a live autocomplete search bar inside the Nexus graph toolbar, enabling users to quickly locate and navigate to any handle in the loaded graph without manually scanning all nodes.

- Designed the autocomplete component layout: a text input with a floating dropdown suggestion list positioned directly below it inside the graph-toolbar panel, styled with the existing CSS variable system (font-mono, color-text, color-bg, color-border) to remain visually consistent with the rest of the Nexus panel.
- Implemented the renderSuggestions(query) function in frontend/js/nexus.js — filters all handle-type nodes from the live graphNodes array, matches by prefix (startsWith) on both handle name and label, bolds the matched prefix inside the suggestion list item using a suggestion-match span, and shows a not-found state when the query has no matches in the current graph.
- Added sub-line metadata to each suggestion item: follower count formatted as 1.2M / 340K using the existing fmtFollowers() helper, giving users instant context to identify the correct handle in a large multi-handle graph without clicking.
- Implemented keyboard navigation: ArrowDown and ArrowUp move the highlighted index through the visible suggestion items with scrollIntoView(), Enter selects the highlighted item or falls back to the raw input value, and Escape closes the dropdown and blurs the input. Highlight state is tracked via a highlightedIndex variable and applied as a CSS class.
- Used mousedown (rather than click) on suggestion items to ensure the handler fires before the input's blur event, preventing the dropdown from closing before the selection is registered — a subtle event ordering issue in browser focus management.
- Added a document-level click listener to close the dropdown when the user clicks anywhere outside the input or list, and wired the Refresh button to re-render suggestions after a graph reload so newly available handles immediately appear in the autocomplete.
- Wrote the CSS rules for .search-suggestions, .suggestion-item, .suggestion-item.highlighted, .suggestion-match, .suggestion-sub, and .suggestion-not-found inside the nexus.html style block, including a max-height with overflow-y: auto to cap the dropdown at a readable height for graphs with many handles.

Day 3 | 24 June 2026 | Handle Picker Internal Filter & Selection State Polish

Day 3 added a keyword filter input inside the handle-picker drawer and fixed the visual selected/unselected state for picker rows, which had been reported as unclear.

- Added a .picker-search-wrap div with a #picker-search-input text input immediately below the picker header and above the handle list in frontend/nexus.html, styled with font-mono, a 1.5px border that highlights on focus, and an italic placeholder.
- Implemented the picker internal filter in setupFilterBar(): on every input event the handler reads the query, iterates all .handle-picker__item elements in the live DOM, sets display:none on items whose handle or display-name does not include the query string, and shows a .picker-no-results message when visibleCount reaches zero with a non-empty query.
- Critically, the filter listener was wired once at IIFE initialisation time rather than inside renderList() — the correct pattern since #picker-search-input is a persistent HTML element that is never destroyed or rebuilt, meaning it only ever needs one listener.
- Added the .is-selected CSS class pattern for picker rows: .handle-picker__item.is-selected sets background to var(--color-text) and color to var(--color-bg), giving a clear high-contrast dark fill to selected rows and making the selection state immediately readable at a glance without requiring close inspection of the checkbox.
- Updated the toggle() callback inside the forEach listener to call item.classList.toggle('is-selected', isNowChecked) in sync with the selectedHandles Set update, ensuring the visual and data states are always consistent with no additional re-render cycle needed.
- Applied the initial .is-selected class on render for handles already in selectedHandles so that closing and re-opening the picker (or clicking Select All then narrowing the date range) preserves the correct visual state on all rows immediately.
- Closed the picker search filter when the picker is closed by clearing the input value, restoring all item visibility, and removing any .picker-no-results message inside the existing closeBtn click handler.

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Day 4 | 25 June 2026 | Critical SPA Layout Bug — Filter Bar Invisible on Navigation

The most significant structural bug of the week was diagnosed and fixed on Day 4: the Nexus filter bar was completely invisible whenever a user navigated to the Nexus page via the SPA router (clicking the nav link from another page), though it appeared correctly on hard/direct page loads.

- Traced the root cause to the SPA router's page-swap mechanism in nav.js: the router identifies the current page's element via document.querySelector('main') and replaces its innerHTML with the fetched page's content. In nexus.html the was nested inside a , so the filter bar, handle picker, and nexus-wrapper div were all outside and were never injected by the SPA swap.
- Fixed by restructuring nexus.html so that is the outermost content container — the filter bar, handle-picker drawer, and graph-split div are all direct children of , meaning the SPA swap correctly injects the entire Nexus UI on every client-side navigation.
- Applied position:fixed to .nexus-page with top:var(--nav-height), left:0, right:0, bottom:0 so the page fills the viewport correctly regardless of what parent element the SPA injects into — eliminating dependency on any specific DOM hierarchy from the previous page.
- Renamed .nexus-container to .nexus-split and updated all CSS references to reflect the new semantics: .nexus-page is the fixed full-viewport outer wrapper, .nexus-split is the flex row containing the D3 canvas and the right panel.
- Changed .handle-picker from position:fixed to position:absolute — this was required because position:fixed descendants are repositioned relative to any ancestor with a CSS transform applied. The SPA entrance animation applies transform:translateY() to during the 500ms transition, which would offset a position:fixed picker. Since .nexus-page itself fills the full viewport via position:fixed, position:absolute on the picker achieves identical visual placement with no transform-containment side effects.

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WEEK 6 — DAILY WORK LOG (continued)

Day 5 | 26 June 2026 | Graph Simulation Timing Fix & Resize Observer

Day 5 addressed the second major structural bug: all graph nodes rendering clustered at the top-left corner of the SVG canvas, making the graph appear blank or broken immediately after page load.

- Identified the root cause: initSimulation() was called synchronously inside initNexus() before the browser had performed layout. At that point svg.node().clientWidth and clientHeight both return 0, so d3.forceCenter(0/2, 0/2) = forceCenter(0, 0) was set. Subsequent simulation ticks pulled all nodes toward the SVG origin (top-left), rendering them outside the visible canvas area.
- Fixed by replacing the synchronous initSimulation() call with a double-nested requestAnimationFrame wrapper: the outer rAF waits for the current paint frame to commit, the inner rAF runs in the following frame when layout dimensions are fully resolved. initSimulation() and loadGraph() are both deferred inside this wrapper.
- Added a forceCenter update inside buildBaseGraph() itself: after reading the actual clientWidth and clientHeight at graph-build time, simulation.force('center', d3.forceCenter(centerX, centerY)) is called explicitly to correct any stale (0, 0) center that may have been set during the early initSimulation() call.
- Implemented initResizeObserver() using the browser's ResizeObserver API to watch the SVG element and update forceCenter whenever the canvas dimensions change — handling window resizes, right-panel expand/collapse, and any future layout shifts that alter the graph area without a full page reload.
- The ResizeObserver is registered with a gentle alpha(0.1).restart() on each resize event rather than a full alpha(1) reset, so nodes re-centre gradually without the explosive spread that a hard reset would produce.
- Registered the ResizeObserver cleanup inside the SPA abort signal listener, calling ro.disconnect() when the signal fires to prevent the observer from continuing to run after the user navigates to a different page.

Day 6 | 27 June 2026 | DOM Scoping Bugs — Tooltip Crash & Second-Search Crash

Day 6 fixed two independent crashes that each manifested only under specific interaction sequences, making them harder to catch in basic testing.

- BUG A — Tooltip null crash on SPA navigation: #node-tooltip and #toast were placed in the HTML body after . The SPA router only injects the element, so on client-side navigation both elements were absent from the DOM. nexus.js stored them as const tooltip = getElementById('node-tooltip'), which became null. On the first node hover, tooltip.innerHTML = ... threw a TypeError, crashing the entire D3 event system and disabling all subsequent node interactions.
- Fixed by moving both and inside the element in nexus.html, immediately before . Since the SPA now injects the full content, both elements are always present in the DOM regardless of navigation path.
- BUG B — Second-search crash from cloneNode/replaceChild anti-pattern: the picker filter listener was being set up inside renderList() using cloneNode(true) and parentNode.replaceChild() to remove old listeners. On the first search this worked, but it detached the original element from the DOM. On the second search, pickerSearchInput.parentNode was null (detached element), throwing 'Cannot read properties of null (reading replaceChild')' and crashing renderList() — leaving the picker blank after every second date-range search.
- Fixed by removing the cloneNode/replaceChild block entirely from renderList() and instead attaching the filter input event listener exactly once at IIFE initialisation time. Since the input is a persistent HTML element that is never rebuilt, one permanent listener is all that is needed — the handler always operates on whatever items are currently in the live pickerList DOM.
- Also moved the pickerSearchInput.value reset to the top of renderList() as a clean two-line block (reset value + restore all item visibility), making the reset logic explicit and independent of the listener setup.

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Day 7 | 28 June 2026 | End-to-End Testing, Regression & Documentation

The final day was spent on comprehensive regression testing across all bug fixes and new features, with documentation updates for the full week's changes.

- Tested the autocomplete search bar across graphs of varying size (2 handles to 18 handles), confirming prefix matching, follower count sub-label display, keyboard navigation, Enter-to-select, Escape-to-close, and not-found messaging all work correctly in every case.
- Verified the picker internal filter correctly hides/shows rows in real-time as the user types, handles partial matches mid-string, and shows the no-results message only when the query is non-empty — confirming the filter does not incorrectly hide all rows on an empty input.
- Confirmed the SPA layout fix works across all navigation paths: Nexus loaded directly (hard load), Nexus navigated to from Overview, Handles, Network, Campaigns, and Scraper pages via the nav bar, and Nexus navigated away from and back to — filter bar, handle picker, graph, and right panel all render correctly in every case.
- Confirmed the double-rAF fix resolves the node clustering: on a 1440×900 viewport with 18 loaded handles, all nodes render correctly spread in a circle at the centre of the SVG canvas on every page load and SPA navigation.
- Performed a full regression pass across all 11 existing pages confirming no regressions from the nexus.html restructure, nav.js download fix, or CSS changes — Overview, Handles, Hashtags, Topics, Network, Campaigns, Tweet Analysis, Scraper, Data Explorer, Intelligence, and Nexus all navigate correctly.
- Updated project documentation to describe the autocomplete component API, the picker filter pattern, the SPA-safe page structure requirements (all content inside), and the double-rAF deferred simulation pattern for future contributors adding D3 visualisations to other pages.

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WEEK 6 — SYSTEM ARCHITECTURE: LOW LEVEL DESIGN (LLD)

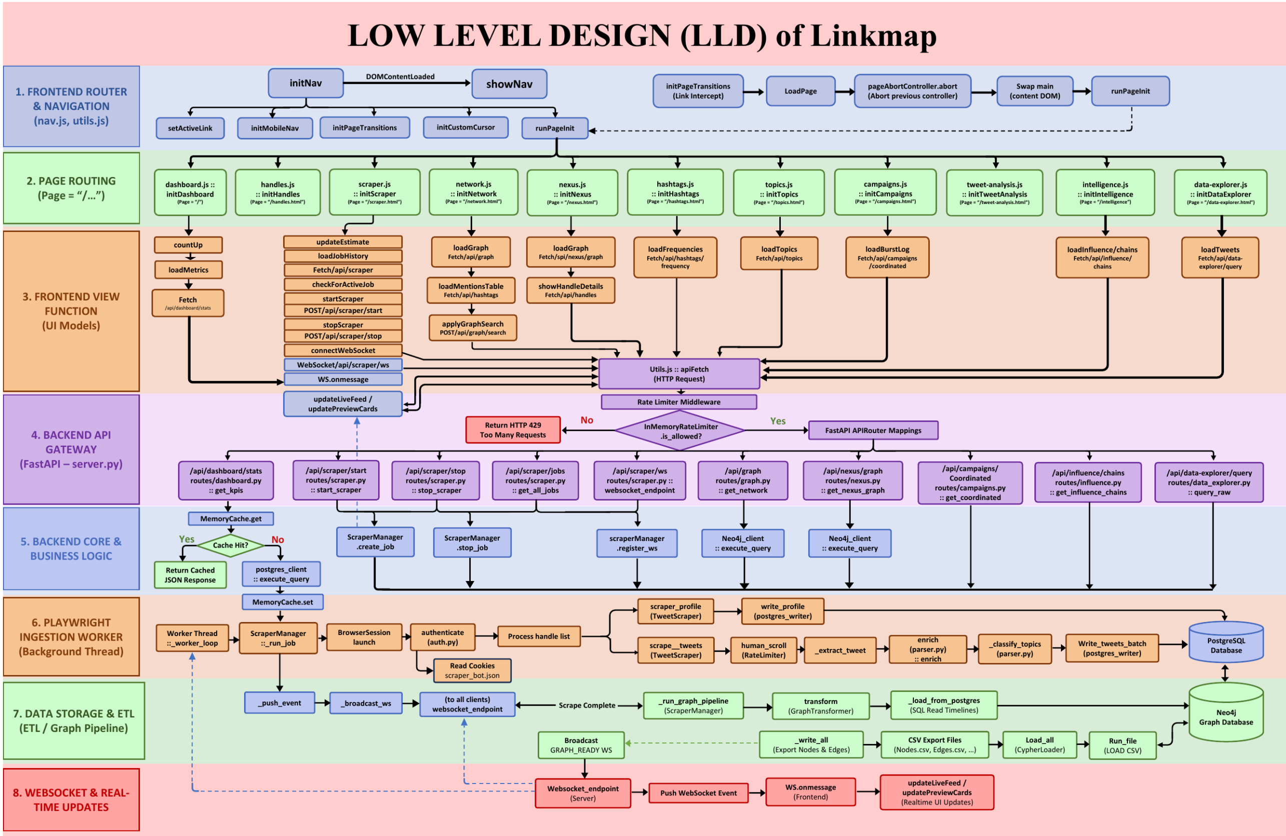

Figure 1: Low Level Design (LLD) of the Linkmap OSINT Platform — complete data flow from frontend SPA router through FastAPI middleware, business logic, Playwright scraper, PostgreSQL storage, Neo4j graph pipeline, and WebSocket real-time update layer.

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WEEK 6 — MEMBER CONTRIBUTIONS

Rishav Pal — Data Engineer

- Led diagnosis of the SPA layout bug — traced the root cause to the element scoping issue in nav.js and nexus.html, confirmed the fix across all 11 navigation paths including hard loads and SPA transitions.
- Implemented the double-rAF deferred simulation pattern to fix the node clustering bug, and wrote the forceCenter update inside buildBaseGraph() to correct stale zero-values from early initialisation.
- Implemented initResizeObserver() with cleanup wired to the SPA abort signal, ensuring the observer is properly disconnected on page navigation.
- Diagnosed the tooltip/toast DOM scoping crash — confirmed that #node-tooltip and #toast being outside caused null references on every SPA navigation, and confirmed the fix.
- Led regression testing of the simulation timing fix across multiple viewport sizes and handle counts.
- Updated documentation covering the double-rAF pattern and SPA-safe page structure requirements for future D3 visualisation pages.

Sayantan Mandal — OSINT Analyst

- Designed and implemented the autocomplete handle search bar in the Nexus graph toolbar: renderSuggestions(), prefix matching against live graphNodes, bolded match highlighting, follower-count sub-labels, and the not-found empty state.
- Implemented full keyboard navigation for the autocomplete: ArrowUp/Down index tracking with scrollIntoView, Enter-to-select with fallback to raw input, and Escape-to-close with blur.
- Diagnosed and fixed the second-search crash caused by the cloneNode/replaceChild anti-pattern inside renderList() — identified the detached-element parentNode null condition and replaced the pattern with a single one-time listener at IIFE initialisation.
- Wrote the CSS block for .search-suggestions, .suggestion-item, .suggestion-item.highlighted, .suggestion-match, .suggestion-sub, and .suggestion-not-found with max-height overflow-y: auto.
- Conducted end-to-end QA of the autocomplete feature across graphs of 2 to 18 handles, testing every interaction path including keyboard navigation, mouse selection, and post-refresh suggestion re-render.
- Updated project documentation describing the autocomplete component architecture and the correct one-time listener attachment pattern.

Shivam Kumar — Graph Analyst

- Built the picker internal filter: added #picker-search-input HTML element and .picker-search-wrap CSS to nexus.html, wired the input listener once at IIFE initialisation, and implemented real-time hide/show of .handle-picker__item rows with a .picker-no-results empty-state message.
- Implemented the .is-selected visual state for picker rows — CSS class toggled synchronously with the selectedHandles Set, giving a high-contrast dark fill to selected rows and applied correctly on initial render and on every toggle.
- Fixed the handle-picker position from position:fixed to position:absolute to eliminate CSS transform containment conflicts during the SPA entrance animation.
- Fixed the #node-tooltip and #toast DOM scoping bug by moving both elements inside the element, resolving the null reference crash that disabled all node hover interactions on SPA navigation.
- Performed full regression testing of all 11 existing pages to confirm no regressions from any structural or CSS changes introduced during the week.
- Updated project documentation to describe the picker filter pattern, the .is-selected CSS class convention, and the position:absolute requirement for drawers inside position:fixed page containers.

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WEEK 6 — TECHNICAL SUMMARY

Feature Overview — Nexus Graph Explorer Stabilisation

Week 6 resolved six distinct bugs across the Nexus page and added two user-facing features. The autocomplete search bar and picker filter together make the Nexus interface usable at scale — with 18 or more handles in the graph, both the graph search and the report picker were previously navigable only by visual scanning. The five bug fixes (SPA layout, simulation timing, tooltip/toast scoping, position:fixed containment, cloneNode crash) were all root-cause fixes rather than workarounds, ensuring the underlying patterns are stable for future additions.

Component	Technology	Detail
Autocomplete search	Vanilla JS + CSS	Prefix match on live graphNodes, keyboard nav (Up/Down/Enter/Esc), mousedown selection, sub-label metadata
Picker internal filter	Vanilla JS + CSS	One-time input listener, real-time display:none toggle on items, no-results state, reset on picker close
Selected-row state	CSS class toggle	.is-selected applied synchronously with Set update; initial state applied on render
SPA layout fix	HTML restructure	All Nexus content moved inside ; .nexus-page uses position:fixed for viewport-independent sizing
Simulation timing fix	Double rAF + ResizeObserver	initSimulation deferred to second rAF frame; forceCenter updated at buildBaseGraph(); ResizeObserver for live resize
Tooltip/toast scoping	HTML element placement	#node-tooltip and #toast moved inside so SPA swap always injects them
Picker position fix	CSS positioning	position:fixed to position:absolute to avoid CSS transform containment during SPA entrance animation
cloneNode crash fix	JS listener pattern	Replaced cloneNode/replaceChild anti-pattern with single one-time listener; renderList() only resets input value

Week 6 Deliverables Summary

Deliverable	File	Status
Autocomplete search bar (HTML + CSS)	frontend/nexus.html	✓ Complete
Autocomplete logic (renderSuggestions, keyboard nav)	frontend/js/nexus.js	✓ Complete
Picker internal filter input (HTML + CSS)	frontend/nexus.html	✓ Complete
Picker filter logic (one-time listener)	frontend/js/nexus.js	✓ Complete
Selected-row visual state (.is-selected)	frontend/nexus.html + .js	✓ Complete
CRITICAL FIX — SPA layout (filter bar invisible)	frontend/nexus.html	✓ Fixed
CRITICAL FIX — Simulation timing (nodes at 0,0)	frontend/js/nexus.js	✓ Fixed
CRITICAL FIX — ResizeObserver for live resize	frontend/js/nexus.js	✓ Fixed
CRITICAL FIX — Tooltip/toast null crash (SPA nav)	frontend/nexus.html	✓ Fixed
CRITICAL FIX — position:fixed picker containment	frontend/nexus.html	✓ Fixed
CRITICAL FIX — cloneNode second-search crash	frontend/js/nexus.js	✓ Fixed
Regression test — all 11 pages	All frontend pages	✓ Passed
Documentation updates	Project docs	✓ Complete

Bug Fix Reference Table

Bug ID	Symptom	Root Cause	Fix
BUG-01	Filter bar invisible on SPA nav	not outermost container; SPA swap missed filter bar	Restructured nexus.html; all content inside ; position:fixed on .nexus-page
BUG-02	Nodes cluster top-left on load	initSimulation() reads clientWidth=0 before layout; forceCenter(0,0)	Double-rAF defer; forceCenter updated at buildBaseGraph()
BUG-03	Tooltip/toast crash on SPA nav	#node-tooltip/#toast outside ; null after SPA swap	Moved both elements inside
BUG-04	Picker offset during page transition	position:fixed child inside CSS-transformed ancestor (SPA animation)	Changed handle-picker to position:absolute
BUG-05	Second date-range search crashes picker	cloneNode detaches element; parentNode=null on second call	Single one-time listener at IIFE init; removed cloneNode
BUG-06	Graph nodes don't re-centre on resize	No mechanism to update forceCenter after SVG resize	ResizeObserver watches SVG; updates forceCenter; alpha(0.1).restart()

Report prepared by: Rishav Pal (Data Engineer), Sayantan Mandal (OSINT Analyst), Shivam Kumar (Graph Analyst) | Week 6: 22 June – 28 June 2026